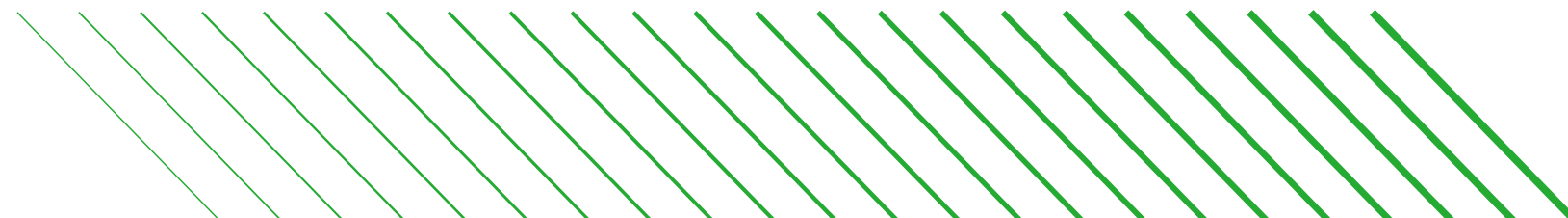


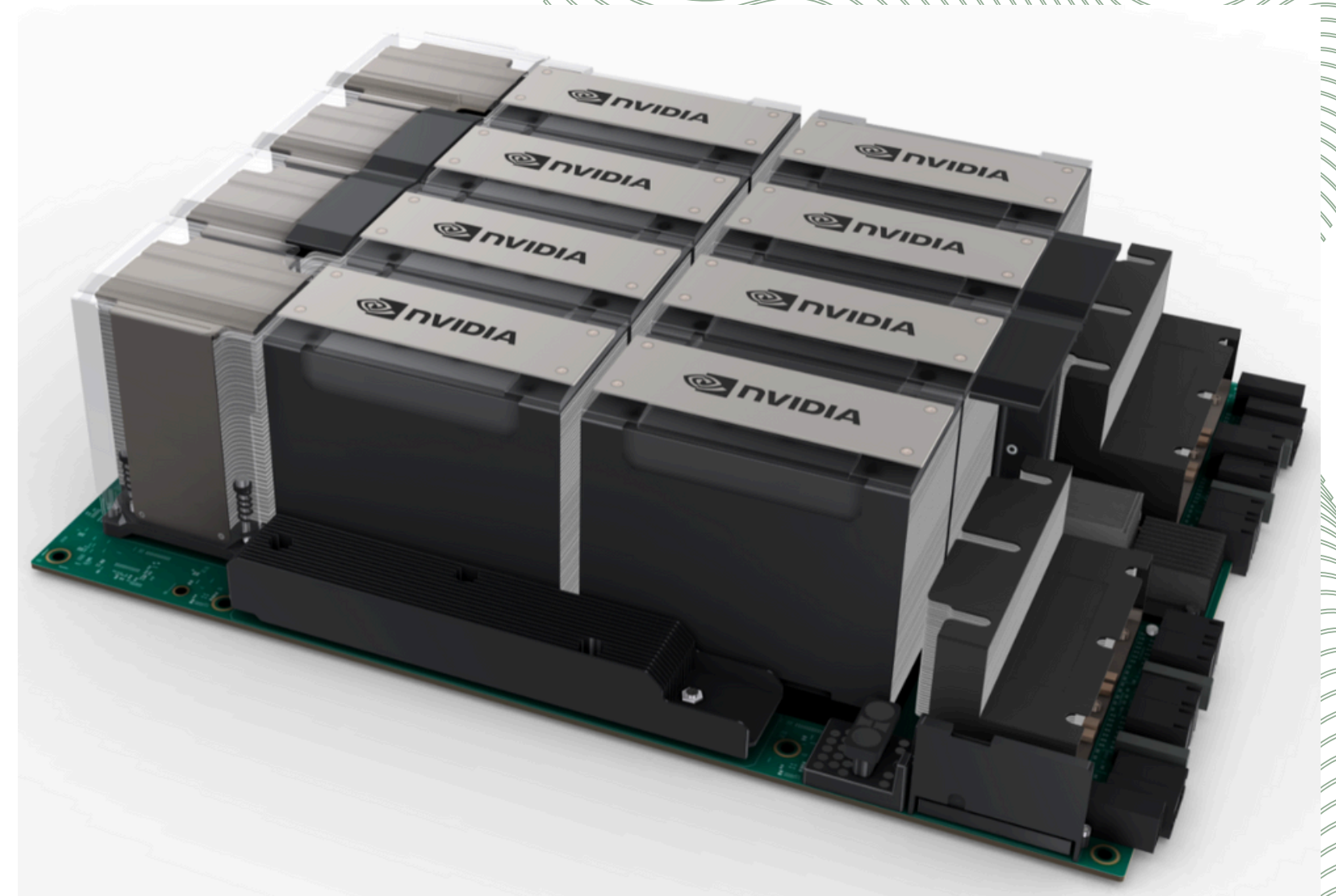


Presented by: Daniel

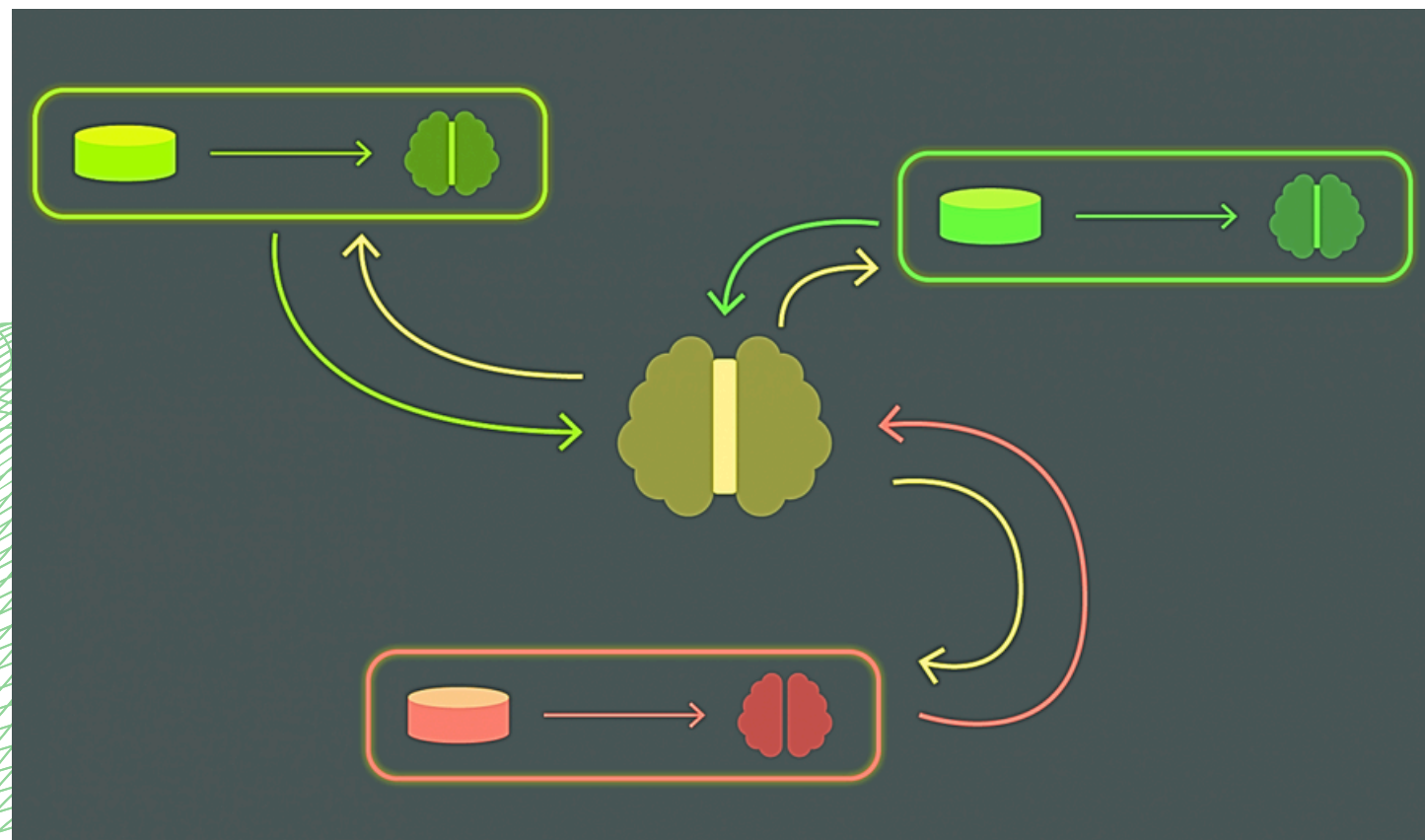


PROBLEM STATEMENT

Large amounts of data remain untapped in machine learning due to privacy concerns. While federated learning addresses this by keeping data with users, it requires significant hardware, making it impractical for users. As a result, valuable data goes unused due to privacy and cost barriers.



SOLUTION



- **Trusted Compute Environments**

Users can choose and verify the training environment

- **No need for Hardware**

Users don't need to acquire expensive compute hardware

- **Easy Payments**

Micropayments enable the use of any amount of data, making it scalable and cost-effective.

- **Privacy-preserving**

Users maintain full control over how their data is used and where it is processed.

FEDLEARN

Privacy-preserving ML Model Training
on Warden.

1

Select ML models to contribute

You can select ML models to contribute your training data on Warden's FedLearn marketplace.

2

Choose your compute provider

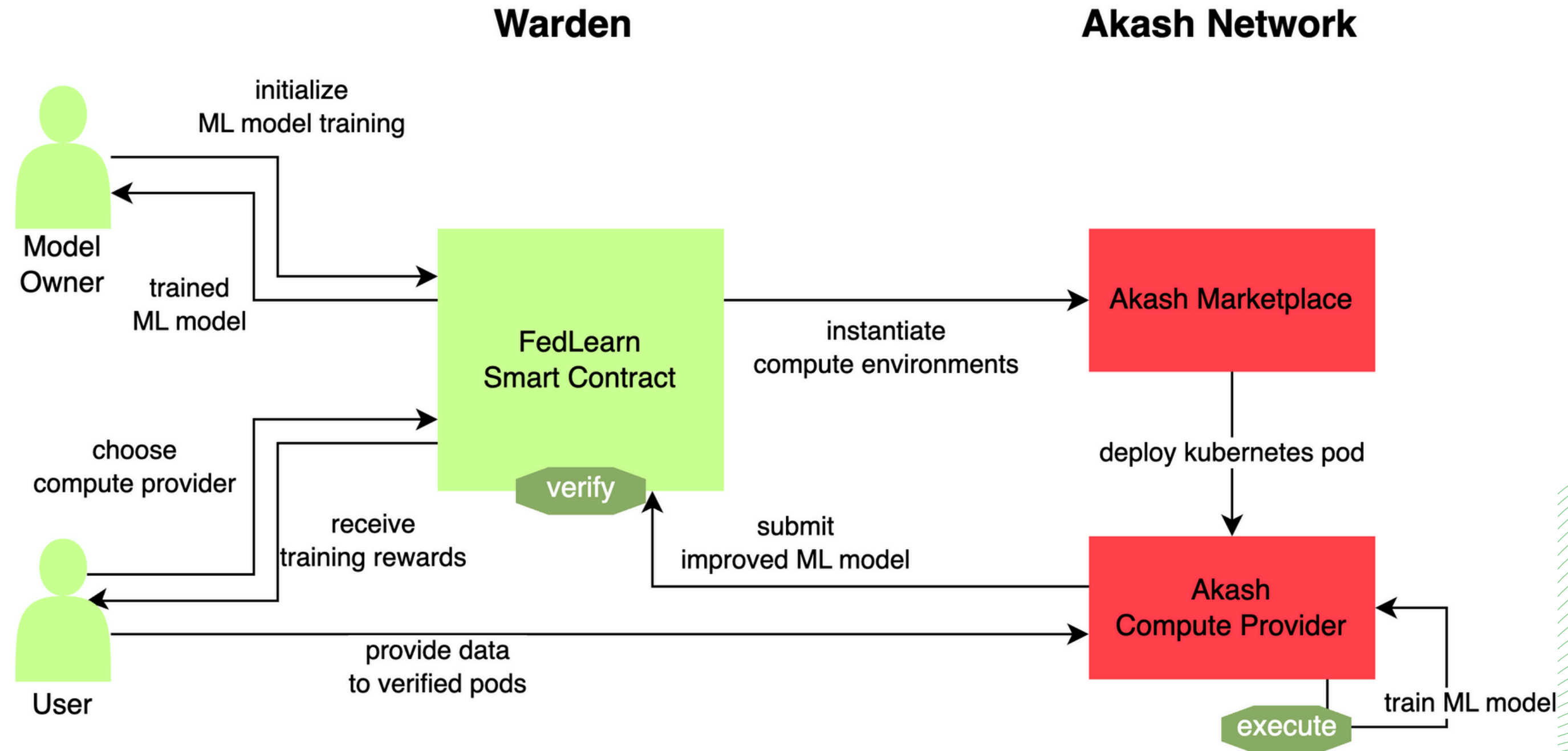
You can choose a compute provider on Akash Network which executes a verified and trusted Kubernetes pod to process your data.

3

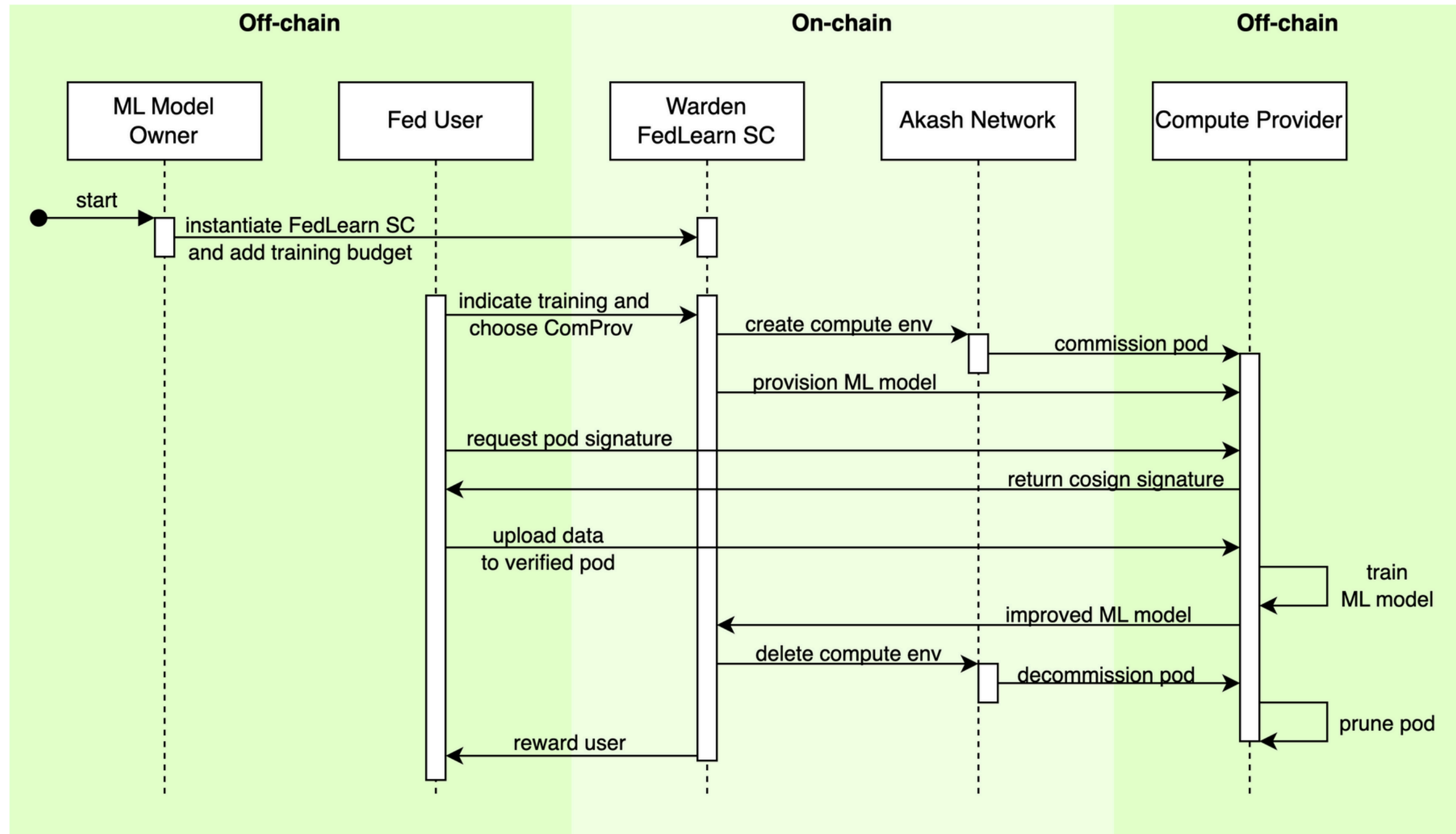
Train ML models and receive rewards

Train ML models in verified compute environments and receive rewards while preserving your privacy.

FEDLEARN



FEDLEARN



WARDEN INTEGRATION

- **x/act**

Using the x/act module, FedLearn executes actions—such as instantiating secure compute instances and managing model updates—on connected chains based on triggers like training initiation, data submission, and task completion. This automation ensures efficient and timely responses to reduce computation costs.

- **x/warden**

The x/warden module is used to manage keys that trigger transactions on the Akash Network, enabling the asynchronous commissioning and decommissioning of compute environments when federated ML model training is initiated or completed.

- **Warden Futures Handler**

FedLearn will leverage Warden Protocol's FUTURES to asynchronously provision compute environments on the Akash network and conduct ML model training. This enables FedLearn to initiate long-running training processes while receiving callbacks upon completion to verify the improvements in the ML models.

FEDLEARN COMPONENTS



- **FedLearn Marketplace**

ML Model Owners can list FedLearning jobs that Fed Users can select to train the ML models and earn rewards.

- **FedLearn Commons**

FedLearn Commons are publicly available base ML models (e.g., LLMs) collaboratively trained by the community, offering ML Model Owners high-quality models and reducing individual training costs.

- **FedLearn Pods**

FedLearn provides open source Kubernetes pods for ML model training that can be verified by cosigns

NEXT STEPS

1

Implement Smart Contracts

Continue implement the FedLearn Marketplace that matches ML Model Owners and Fed Users.

2

Create Trusted Pods

Create open source kubernetes pods that can be used by ML Model Owners and Fed Users to train ML models.

3

Onboard Users and Collaborate

Onboard the first ML Model Owners and Fed Users to FedLearn.